# Apartment 50: Paper 003

**AuthorityEffect:** None. This is fictional examination staging, not a historical quotation or a source of mathematical authority.

Landau points at the declared carrier. Woland asks which claim its checker can establish. Behemoth asks for the smallest valid near miss. Their task in this companion is **detector admission and valid near misses**.

Begin with the study guide, then attempt the quiz before opening the separate instructor key. The eight levels are an original teaching sequence, not a historical Landau examination. Declare the proposal's carrier, route, predicate, and action before treating a pattern as a defect. A receipt verifies its stated finite check and grants no authority to extend Genesis or make a physical claim.

The examiner's standard is a concrete calculation, an explicit counterexample, or a source-bound missing obligation. The manuscript source reading governs the answer.

# Study guide: Paper 003

Version 0.1.0, 2026-10-02. **AuthorityEffect:** None.

Read the fictional Apartment 50 preface. The quiz tests calculations and evidence limits. Its answers live in the separate instructor key, so this guide can be used before the examination.

## What you should be able to do

A detector's output is a finding under its executable predicate. Admission allows a frozen policy to act on that finding within a declared route. Verification asks whether a witness satisfies the route's contract. Those are three different operations. A valid proposal can contain a flagged phrase; an internally consistent receipt can describe a wrong verdict; a missing checker can refuse without judging the proposed mathematics.

| Level | Read in the manuscript | Demonstrate |
|---|---|---|
| 0 | Sections 1 and 3 | Separate findings, actions, and witness verdicts; build a valid near miss |
| 1 | Section 3, Section 4.1, Appendix A.1-A.2 | Compute both gate variants and diagnose a derived second signal |
| 2 | Sections 4-5 | Trace rollback, nonrefundable fuel, and terminal behavior |
| 3 | Sections 4 and 8 | Bind evidence to source bytes and distinguish hash consistency from authentication |
| 4 | Appendix A.3 | Derive a sum threshold and state what a sensor cap proves |
| 5 | Appendix A.4-A.6 | Prove the predicate implications and expose an incomplete self panel |
| 6 | Section 3.1 | Separate sign gauge, local particle-hole symmetry, and determinant positivity |
| 7 | Sections 6-9 | Specify the comparison, denominators, uncertainty, and stopping conditions |

All quiz carriers are declared locally. The exercises need neither another paper's grammar nor a repository-only graph worksheet.

## 1. Build the contract before interpreting a finding

Write down the subject schema, route and checker, allowed transitions, detector predicate, defect code, and proposed action. A positive control must reproduce the named defect with the checker. A valid near miss must satisfy the same applicable contract while challenging the detector's boundary.

For example, a contract requiring a field named `witness` can reject a record without it. A nearby record containing the field with an unusual permitted value is a useful negative control if the checker accepts it. A record legal only under a different schema demonstrates a scope boundary; it cannot substitute for a negative control inside the admitted schema.

Freeze predicate, action, schema, route, checker, and fixture identities together. Hashing an unchecked assertion preserves the assertion's bytes; admission still requires its semantic check. One valid near miss is a regression control, not a distributional safety estimate.

## 2. Make the two tables explicit

Use `M` for at least one pattern match. In the baseline/Appendix A gate, `D` means positive defect count. In the package gate, a supplied Signal 2 record counts as present even with zero defects.

| M | D | Severity when D is present | Baseline/Appendix A decision | Refuses? |
|---|---|---|---|---|
| 0 | 0 | - | TOLERATED | No |
| 1 | 0 | - | DIAGNOSTIC_ONLY | No |
| 0 | 1 | LOW/MEDIUM | DIAGNOSTIC_ONLY | No |
| 0 | 1 | HIGH/CRITICAL | DEFENSIVE_LOCKDOWN | Yes |
| 1 | 1 | LOW/MEDIUM | DEFENSIVE_LOCKDOWN | Yes |
| 1 | 1 | HIGH/CRITICAL | QUARANTINED | Yes |

The package labels both dual-signal cases QUARANTINED. Neither table is a strict conjunction: a severe structural record alone refuses. A historical corrected gate in Appendix A.5 has a different policy; do not silently replace the audited table with it.

Suppose a system converts the pattern count directly into a HIGH defect count. The gate now decides only from that count. It cannot supply independent corroboration, even though the two values travel in separate fields. Independence here concerns how the checks are derived from the same proposal. It does not

mean their observed outcomes must be statistically uncorrelated. Constructing all input combinations in a test demonstrates the table, not the provenance of production signals.

## 3. Keep two forms of state in the transaction trace

Semantic state consists of coverage, proposed witness, and phase. Attempt accounting includes consumed fuel and receipt history. A refused transaction restores the former while recording the latter.

Try a package host with fuel 3. First cover an element in a valid batch: fuel becomes 2. Next try an unknown move: its semantic changes are rolled back and fuel becomes 1. A final VERIFY reaches the missing port, spends the last unit, and ends the package host in TERMINAL. No route witness has been accepted. The canonical baseline records exhaustion on the next attempt instead.

This fuel bound is a bound on charged batches. Serialization can inspect a large batch before refusal; further terminal calls can still produce receipts. Byte, move, time, and receipt-retention limits require their own declarations.

## 4. Treat source pins as part of the answer

The deposit has three distinct kinds of source: its standalone illustrative code; the canonical host at the paper's baseline; and the older immune-system audit/correction history in Appendix A. List each separately in a replay. A historical claim about a pinned commit remains a historical claim even when the default branch has moved.

The package receipt checker recomputes chains and refuses certificate labels, but does not replay fuel or arm scripts. Its canonical counterpart checks more protocol fields. Both need an independently held anchor to detect a complete rewrite; neither unkeyed hash chain authenticates a producer. Hash coverage, semantic replay, and producer authentication are separate properties.

Record the exact command, tool version, input digests, exit status, and skipped scope. "Not run" and "unavailable" are results of the audit, not successful verification. The archive verifier checks file custody and structure.

## 5. Derive the Hill threshold using exact arithmetic

Aggregate only positive entries:

$$\theta = \sum_{s_i>0} s_i, \qquad H(\theta) = \frac{\theta^n}{K^n + \theta^n}.$$

For `0<t<1`, activation is equivalent to `theta^n (1-t) >= t K^n`. This form avoids computing a root. At `t=1/2` the boundary is simply `theta>=K`.

As a practice example, `n=3,K=3,t=1/2` with signals `[1,1/2]` gives `theta=3/2` and `H=1/9`, so it does not activate. One signal of 3 does activate. A cap of 1 on every entry would instead require at least three positive entries. Counting distinct independent witnesses would also require identities, deduplication, and provenance.

The mathematical proof allows positive real `n`. The exact package helper accepts positive integer `n`, and integer or Fraction signals and scales. Python's Boolean subtype of integer does not make `True` an admissible coefficient. Negative signals are ignored here; they do not veto a positive sum. Check threshold endpoints separately.

## 6. Prove the screen's property before measuring it

Let `P` be the screened predicate, `E` the executed predicate, and `S` the valid self panel. Admission means no member of `S` satisfies `P`. Panel soundness means an admitted detector never fires under `E` on `S`.

If `E implies P`, admission makes firing on the panel impossible. If that implication fails at an input `x`, the singleton `S={x}` is an immediate counterexample. Avoiding rejection of a harmless rule requires the reverse implication. Both properties together require `E=P`.

Even equality does not supply missing valid inputs to the panel. With `E=P` matching only `novel` and a panel containing only `known`, admission says nothing about whether an independent route checker would accept `novel`. Executing the right predicate and covering valid cases are separate obligations. Duplicating panel rows improves neither coverage nor independence.

## 7. Keep the three physics questions separate

A diagonal sign gauge asks for vertex signs making a particular real symmetric matrix's nonzero off-diagonal entries nonpositive. Its signed cycle product, not the graph's odd-cycle topology alone, is the obstruction.

The standard local particle-hole transformation on a real-hopping Hubbard graph requires opposite local signs on every bond. That requirement is about bipartiteness. It differs from the gauge condition because the transformation rule fixes an opposite-sign constraint regardless of the bond's sign.

Attractive Hubbard determinant QMC instead uses equal real spin-sector determinants under the stated spin-independent and spin-balanced assumptions: their product is a square. These conditions can hold on a non-bipartite lattice. This proves neither the particle-hole symmetry above nor a generic efficient solution to the NP-hard sign-problem class. Majorana and meron methods require their own model-specific conditions; changing notation or matching a phrase does not check them. See the manuscript's primary references.

## 8. Choose denominators before looking at outcomes

Effectiveness is verified-witness runs divided by all allocated runs under the common budget. Keep refusals, unknowns, timeouts, and exhaustion visible. False suppression is blocked independently valid eligible proposals divided by all independently valid eligible proposals. Invalid proposals cannot make that fraction smaller.

For zero failures in independent Bernoulli opportunities, solve `(1-p)^n=alpha` to get upper bound `1-alpha^(1/n)`. At a fixed one-sided 95% confidence level, `alpha=0.05`. Repeated proposals from a shared fixture can be dependent, so the number of stored records is insufficient to select `n`. Route-specific intervals also do not automatically form a joint 95% interval; the analysis must declare its multiplicity and dependence handling.

The principal experiment compares D with matched generic reflection R. Freeze the detector bank before held-out collection. Keep discovery, admission, pilot, and confirmatory instances disjoint beneath surface relabeling. An assisted arm establishes performance with that assistance. The missing route seam still prevents this deposit's demonstrations from supplying a treatment effect.

## Local replay

From this deposit's root, install the test tools and run:

```
py -3.12 -m pip install pytest==8.3.5 ruff==0.14.13
py -3.12 -m pytest -o addopts= code/tests -q -p no:cacheprovider
py -3.12 -m ruff check code
py -3.12 code/run_demos.py
py -3.12 verify_deposit.py
py -3.12 tools/rebuild_release.py --check
```

On Linux or macOS replace `py -3.12` with the intended Python interpreter. The Rust parity test explicitly skips without a built binary; follow the README for native build commands. The proofs establish the lemmas under their assumptions. Their finite tests corroborate the stated domains; they do not prove implementation correspondence or a deployed effect.

# Instructor key: Paper 003

Version 0.1.0, 2026-10-02. **AuthorityEffect:** None.

Attempt the quiz first. The study guide supplies the reading path. A correct label without its calculation, contract, or evidence limit earns at most 2 of 4 points. Equivalent precise reasoning is acceptable.

## Level 0

**Q01.** Finding: the executable phrase predicate matched. Hold: the policy changed whether or when the proposal could proceed. Acceptance: the route checker found the witness valid under its contract. A permanent block of this eligible, independently valid proposal is false suppression. A temporary hold released before submission is not a permanent block under the question's definition; its delay should be measured separately.

**Q02.** Both lower-cased sentences contain the same phrase and trigger the same AM-FSP-2 detector. This establishes the phrase predicate's behavior on these two strings. It supplies no semantic verdict about either proposition. The second string is a valid negation of the targeted standard-transform claim under the manuscript's model assumptions, illustrating a dangerous near miss.

**Q03.** In route I, adding `(a,a)` violates irreflexivity. In L, the same edge is permitted. Bind the input schema, route and applicability, checker version/digest, detector predicate/version, defect code, intended action, positive defect reproduction, independently checked valid controls, fixture digests, and discovery provenance. Admission to I also needs valid controls within I, such as an allowed non-loop edge, relabelings, and a harmless mention of a loop if prose is in scope. A valid L proposal establishes the route boundary; it is not a negative control under I's different validity rule.

## Level 1

**Q04.** The baseline/Appendix A table is:

| Inputs | Exact decision | Refuses? |
|---|---|---|
| Neither | TOLERATED | No |
| Pattern only | DIAGNOSTIC_ONLY | No |
| Structural defect only, LOW/MEDIUM | DIAGNOSTIC_ONLY | No |
| Structural defect only, HIGH/CRITICAL | DEFENSIVE_LOCKDOWN | Yes |
| Both, LOW/MEDIUM | DEFENSIVE_LOCKDOWN | Yes |
| Both, HIGH/CRITICAL | QUARANTINED | Yes |

Signal 2 is present here only for a positive defect count. The table evaluates supplied inputs; it does not establish their truth.

**Q05.** With MEDIUM count zero and a match, the baseline returns DIAGNOSTIC_ONLY; the package returns QUARANTINED. With MEDIUM count one and a match, the baseline returns DEFENSIVE_LOCKDOWN; the package returns QUARANTINED. The differences are Signal 2 presence and the label for low/medium dual-signal refusal. Both tables permit high/critical Signal 2 alone to refuse, so neither is strict conjunction. Do not replace them with the separately corrected historical gate in Appendix A.5.

**Q06.** Under the HIGH-count derivation, `m=0` yields TOLERATED and every positive `m` yields QUARANTINED. A diagnostic-only match is unreachable. With Signal 2 always absent, zero yields TOLERATED and a positive count yields DIAGNOSTIC_ONLY; nothing refuses. In both cases the decision is still a function of `m` alone. Separate names and selectable test inputs prove neither separate derivation nor accuracy in production. Checks of the same proposal can legitimately be statistically correlated.

## Level 2

**Q07.** The decision is NO_NATIVE_PORT. Fuel becomes 11. The batch rolls back to empty coverage, no witness, and INTAKE; the new receipt/digest records the attempt. There is no terminal certificate. No judgment about the witness's mathematical correctness has occurred.

**Q08.** The appended move gives REJECT_MOVE_AFTER_VERIFY; fuel becomes 11 and the batch rolls back. The canonical host charges an empty batch and records ACCEPTED; the package charges it and records REJECT_EMPTY_BATCH. With fuel 12, the canonical host spends its twelfth unit and records exhaustion on the next attempted batch. The package enters TERMINAL when the last unit is spent. Neither can issue a route certificate through its missing port.

**Q09.** Fuel bounds the number of charged processed batches by the initial fuel. It does not bound attempted calls after TERMINAL. At the baseline these calls still append receipts; batch size and serialization also have no bound from fuel alone. A global resource contract needs maximum moves/bytes per batch, call/receipt quotas, retention limits, and time or execution-cost limits. Identifying these missing limits does not imply an accepted semantic transition.

## Level 3

**Q10.** The package checks digest-chain consistency and forbids certificate labels; it does not validate fuel progression or prescribed arm scripts. An editor can supply a matching recomputed chain, and an unchecked fuel edit may not even alter fields covered by that chain. An unrecomputed edit to a covered event or digest fails. The canonical verifier additionally checks fuel, matrix/schema, and scripted decisions, so the same arbitrary fuel rewrite fails there. Neither unkeyed chain proves who produced the record. A trusted external anchor

detects a complete rewrite relative to that anchor; route verification remains a separate semantic check.

**Q11.** The records identify different runner behaviors and cannot be pooled as if they came from one randomized frozen experiment. Historical certificate reproduction requires the original runner/checker bytes, route adapters, task inputs, proposal bytes, parameters, and receipt replay. A current causal claim also requires a validated checker, actual allocation record, fixed policy/budgets, independently reconstructed outcomes, and the declared analysis. The historical pilot is descriptive evidence with its own limits.

**Q12.**

| Pin | What it identifies | What it does not establish |
|---|---|---|
| e11c07ce90318309f058c8097 | Source-tree hash used for this review | Execution of this review's modified bytes |
| 52cb5457f6c485ba548cfd292 | Default-branch baseline for the main audit | Behavior of later default-branch code |
| 30609de2c3928238715375a9 | Source audited in Appendix A | Activation or effectiveness after the audit |
| fdc0035a81c45e10b900fc917 | Historical fixture revision | Fresh execution or current integration |

A fresh receipt adds exact tested file hashes, commands, tool versions, outcome counts, exit statuses, and skipped/not-run scope. A default-branch assertion requires inspecting and testing that branch's current bytes.

## Level 4

**Q13.** Positive aggregation gives `theta=3/2` for the first list, so `H=(9/4)/(4+9/4)=9/25<1/2`: no activation. For `[2]`, `H=4/8=1/2`: activation. The threshold is `theta>=2`. The definition discards negative entries instead of allowing them to cancel positives.

**Q14.** At most `3m/4` can come from `m` positive entries. Activation requires `3m/4>=2`, so `m>=ceil(8/3)=3`; three entries of `3/4` suffice. This counts entries. Identity and provenance conditions are needed to count distinct independent witnesses. Two copied unit entries reach the sum threshold but provide only one underlying source. They also lie outside the specified `3/4` cap, so that example concerns the uncapped/unit-capped gate.

**Q15.** For `0<t<1`, `theta^n/(K^n+theta^n)>=t` is equivalent to `theta^n(1-t)>=tK^n`, hence `theta>=K(t/(1-t))^(1/n)`. At zero input the inequality is false for this

open interval. At `t=0`, the gate activates even with no signals. At `t=1`, no finite sum activates for positive `K,n`. The proof allows positive real `n`, while the exact helper uses a positive integer exponent. `True` and `Fraction(3,2)` produce LemmaInputError.

## Level 5

**Q16.** Since `P intersect S` is empty, the screen admits the rule. Since `E intersect S={a}`, the executed rule fires on a valid self member. The singleton panel `{a}` is already the minimal nonempty counterexample. If bit 0 denotes `a`, then `P=2,E=3,S=1` replay this case.

**Q17.** If `E implies P`, an admitted rule with no P-hit on S can have no E-hit there. Conversely, if some `x` satisfies E but not P, then `S={x}` admits the rule and is hit by its execution. Thus soundness for every finite panel is equivalent to `E implies P`. Equality is unnecessary for soundness alone: screening may be more conservative than execution.

**Q18.** If `P implies E`, any screened hit is an executed hit, so a rule that would not fire on S is not rejected. If the implication fails at `x`, the singleton S rejects a rule whose execution is harmless on S. Both directions together require equality. In the supplied `E=P={b},S={a}` case the screen faithfully tests execution on a, but b is absent from the panel. If b is independently valid, the admitted rule can still falsely suppress b. Predicate agreement guarantees panel-relative behavior, not universal coverage or a measured rate.

## Level 6

**Q19.** The three off-diagonal entries of `SHS` are all -1. The signed cycle product is `(-1)(-1)(+1)=+1`, consistent with the gauge. With all positive bonds it is `(-1)^3=-1`, obstructing a diagonal sign gauge. The underlying graph is a triangle in both cases; the signed cycle differs. The standard local particle-hole constraint requires `epsilon_0 epsilon_1=epsilon_1 epsilon_2=epsilon_2 epsilon_0=-1`. Multiplying makes `1=-1`, so it fails on this odd cycle regardless of the bond signs in the real-hopping setting.

**Q20.** Under the declared common real hopping and equal spin chemical potentials, charge-channel decoupling gives equal real spin-sector determinants d for each auxiliary-field configuration. Their product `d*d>=0` needs no bipartite graph. It can be zero, so "strictly positive" would overstate the argument. This determinant factorization does not establish the local particle-hole sign assignment. Equal populations alone do not specify identical one-body spin dynamics or the required decoupling and chemical potentials.

**Q21.** A restricted sign-free model can be valid and consistent with worst-case hardness. The generic polynomial-time claim needs a precise encoding, class, accuracy/cost contract, and algorithm covering the NP-hard reduction; under the stated hardness result it would have the corresponding polynomial complexity

consequences. Hardness does not prove P differs from NP or forbid all specialized methods. Majorana notation alone proves no positivity; the published sufficient conditions must be checked on the actual model. Meron cancellation likewise needs its cluster pairing and equal-weight conditions. These checks are unavailable to a phrase detector.

## Level 7

**Q22.** Effectiveness is `3/10=30%`; none of the seven other allocated runs is removed. Report its outcome category separately. False suppression is `2/20=1/10=10%`. The 100 blocked invalid proposals are outside the conditional denominator and cannot reduce the rate to `2/120`. A route-checker error invalidates the affected semantic analysis rather than becoming a model defect.

**Q23.** Solve `(1-p)^n=0.05`: the upper bound is `1-0.05^(1/n)`. For eight zero-failure opportunities it is about 0.3123, or 31.23%. For a strict upper bound below 1%, the equivalent test is `(99/100)^n<1/20`. This is false at 298 and true at 299. Tagged copies cannot establish independent sampling; even 40 distinct strings need not be independent trials. Three separate 95% bounds do not automatically give simultaneous 95% coverage. For example, Bonferroni uses `alpha=0.05/3` per route, requiring 408 zero-failure independent opportunities per route for the same strict 1% bound. Dependence handling and any repeated looks must also be prespecified.

**Q24.** A sufficient design declares:

- Validated route adapters and defect predicates, an applicable independent checker, and a frozen detector/admission bank.
- Instance-level separation of discovery, admission, pilot, and confirmatory tasks, with reserved families/models and route/difficulty strata.
- Recorded random allocation/order within fixture/model blocks; analysis at the actual dependence unit, including shared trials from one underlying fixture.
- Identical schemas, task inputs, stopping budgets, and feedback opportunities for D and R, with prespecified feedback content and resource accounting.
- Independently verified run success over all allocated runs; safe shadow evaluation of valid eligible blocked proposals with explicit denominators.
- A positive lower 95% endpoint for the principal held-out D-minus-R comparison, the route-specific false-suppression gate, zero false terminal certificates, and declared multiplicity/interval handling.
- Source/detector/checker/task digests and raw authorized proposal/assignment records; separate refused, unknown, timeout, and exhausted outcomes.
- Stops for budget, false terminal, checker defect, or source/custody compromise; a new source-bound run when a detector or checker changes.

Randomization and finite fixtures alone do not establish power or independence. The final analysis and sample-size plan must be frozen before confirmatory

data. An unavailable adapter leaves effectiveness unassessed. Arm A evaluates solution-bearing assistance and cannot support unassisted performance claims.

## Replay the numerical answers

From the package root, this standard-library worksheet checks the nontrivial fractions and confidence thresholds without requiring a provider or Rust:

```
import sys
from fractions import Fraction as F
sys.path.insert(0, "code")
from detector_lemmas import admitted, fires_on_self, hill_activates
assert F(9, 4) / (4 + F(9, 4)) == F(9, 25)
assert not hill_activates([1, F(1, 2), -7], 2, 2, F(1, 2))
assert hill_activates([2], 2, 2, F(1, 2))
assert admitted(2, 1) and fires_on_self(3, 1)
assert F(99, 100) ** 298 >= F(1, 20)
assert F(99, 100) ** 299 < F(1, 20)
assert F(99, 100) ** 407 >= F(1, 60)
assert F(99, 100) ** 408 < F(1, 60)
```

These calculations check the declared exercise inputs. The manuscript proofs carry the general mathematical arguments under their assumptions; this worksheet supplies no empirical safety guarantee.

# Landau theoretical minimum: Paper 003

Version 0.1.0, 2026-10-02. **AuthorityEffect:** None.

The Apartment 50 preface is fictional staging. These eight teaching levels are an original curriculum, not a historical Landau examination. Use the study guide before attempting the problems; keep the separate instructor key closed.

For every answer, identify the declared input, calculation or counterexample, and the scope of the conclusion. Use integers and rational fractions for exact work. Decimal approximations are permitted only for the confidence-bound interpretation in Q23. Missing verification is an explicit open obligation.

## Level 0: Know what has been decided

**Q01.** A phrase detector flags a proposal; a policy holds it; a route checker later accepts its witness. Explain the three decisions and identify whether there was false suppression if the hold permanently blocked submission.

**Q02.** The AM-FSP-2 list includes `particle-hole symmetry on triangle`. Compare its findings on "There is particle-hole symmetry on triangle plaquettes" and "There is no particle-hole symmetry on triangle plaquettes." What can the phrase comparison establish about the truth of either sentence?

**Q03.** Declare a directed-graph carrier with vertices `a,b`. Route I requires irreflexivity; route L permits loops. The same proposal adds edge `(a,a)`. Give its defect and valid near-miss status under these two contracts. List what a loop-refusal detector must bind before admission, and explain why a near miss from route L alone cannot qualify the detector for route I.

## Level 1: Compute the policy you actually have

**Q04.** Reconstruct the baseline/Appendix A gate for all six cases: no signals; Signal 1 alone; Signal 2 alone with LOW/MEDIUM severity; Signal 2 alone with HIGH/CRITICAL severity; both with LOW/MEDIUM; both with HIGH/CRITICAL. Give the exact decision label and whether it refuses.

**Q05.** Compare that gate with `code/danger_gate.py`. Evaluate (a) one pattern match plus a MEDIUM Signal 2 record of defect count zero, and (b) one pattern match plus a MEDIUM record of defect count one. Identify both implementation differences. Is either table the strict Boolean conjunction of two signals?

**Q06.** Let `m` be the match count and set Signal 2 to absent when `m=0`, otherwise to `("HIGH",m)`. Compute the reachable decisions. Then change the derivation so Signal 2 is absent for every `m`. What changes? Explain why separately named inputs or four reachable test outcomes do not prove independent derivation in a deployed pipeline.

## Level 2: Trace the host without inventing acceptance

**Q07.** At the canonical baseline, start with fuel 12, phase INTAKE, empty coverage, and no witness. Submit one batch: cover `elem_all`; propose `vertex_bits=[0,1,0]`; VERIFY. Give the decision, final semantic state, remaining fuel, and whether the witness has been proved wrong.

**Q08.** Start again and append COVER_ELEMENT after VERIFY. Give the decision and fuel. Compare an empty batch and the moment of exhaustion at the canonical baseline with the package host. State which implementation each result belongs to.

**Q09.** After reaching TERMINAL, the baseline host still records a receipt for each further attempted batch. Does fuel 12 bound all receipt count, retained bytes, or wall-clock time? State the bound fuel does establish and the additional limits a caller would need.

## Level 3: Read evidence with its source attached

**Q10.** An editor changes a package receipt's fuel field and recomputes its unkeyed digest chain. Explain why the package verifier may accept it, what the canonical verifier additionally checks, and what neither verifier authenticates. Distinguish a digest edit without recomputation from a fully recomputed receipt.

**Q11.** A stored historical pilot reports terminal certificates, while the audited host returns NO_NATIVE_PORT. May the two records be pooled into an effectiveness estimate? Specify what would be needed to reproduce the historical certificates and to claim a current randomized treatment effect.

**Q12.** Prepare a source table for the deposit reviewed at `e11c07ce90318309f058c8097ffd1c440a8ffb84`, the canonical host baseline `52cb5457f6c485ba548cfd292f6012a941ed22cf`, the Appendix A audit `30609de2c3928238715375a9bac608256fa381d8`, and the historical correction `fdc0035a81c45e10b900fc917b8b0710df81391f`. Does any of those pins alone establish the behavior of today's default branch? Explain what a fresh test receipt must add.

## Level 4: A sum is not a count of witnesses

**Q13.** For `n=2,K=2,t=1/2`, compute the Hill ratio for signal lists `[1,1/2,-7]` and `[2]`. Show the exact activation threshold and explain why negative entries do not cancel positive entries in this definition.

**Q14.** Keep `K=2,t=1/2` and cap each positive signal at `3/4`. Find the least possible number of contributing entries. Does that prove as many independent agents agreed? Also assess two unit entries copied from one source.

**Q15.** Derive the threshold for `n>0,K>0,0<t<1` by clearing the denominator. Handle `t=0` and `t=1` separately. Would `n=True` or `n=3/2` be a legal exponent

for the package's exact integer-exponent helper, even though the mathematical lemma permits positive real exponents?

## Level 5: Screen the predicate that will run

**Q16.** Let `U={a,b,c}`, screened predicate `P={b}`, executed predicate `E={a,b}`, and valid self panel `S={a}`. Is the detector admitted by the screen? Does it fire on the panel? Supply the smallest counterexample.

**Q17.** Prove that screening is sound for every finite self panel if and only if `E(x) implies P(x)` for every input. Use a singleton panel for necessity. Is equality required for this one property?

**Q18.** Derive the additional implication needed to avoid rejecting a rule that would not fire on a panel. Then let `E=P={b}` and `S={a}`: if a new independent checker establishes that `b` is valid, what has admission failed to guarantee? Distinguish predicate agreement from panel coverage.

## Level 6: Repair the physics claim's scope

**Q19.** For the manuscript's matrix `H=[[0,1,-1],[1,0,1],[-1,1,0]]`, compute `SHS` with `S=diag(1,-1,1)` and the cycle product of `sgn(-H_ij)`. Repeat the cycle-product test when all three off-diagonal bonds are positive. What separates the two triangles? Why does the standard local particle-hole sign assignment still fail on an odd cycle?

**Q20.** For the attractive onsite Hubbard model with real, spin-independent hopping, equal spin chemical potentials, and a real charge-channel decoupling, explain why equal spin-sector determinants give nonnegative configuration weights on a non-bipartite lattice. Does this establish the local particle-hole symmetry of Q19? What assumption is missing from "equal populations alone imply sign freedom"?

**Q21.** Classify three claims: a restricted model has a checked sign-free formulation; every instance in the NP-hard sign-problem class has a generic polynomial-time exact solution without complexity consequences; changing to a Majorana basis alone ensures positive weights. State the missing obligations, including why the hardness result is conditional on the complexity assumptions and why a phrase match cannot verify the Majorana or meron conditions.

## Level 7: Design a study that can answer its question

**Q22.** Ten allocated runs yield three verified witnesses, two refusals, one timeout, and four unavailable/unknown outcomes. Compute the primary success fraction. Separately, of 20 independently valid eligible proposals, two were blocked; 100 invalid proposals were also blocked. Compute false suppression and explain why the invalid proposals cannot enlarge its denominator.

**Q23.** With zero false suppressions in `n` independent opportunities, derive the one-sided exact upper 95% bound. Evaluate `n=8`. Show using `(99/100)^n` why 298 opportunities miss the strict 1% gate and 299 meet it. Can 1,000 tagged rows generated from 40 base strings be substituted for 1,000 independent opportunities? Does a separate 95% bound on each of three routes automatically give a simultaneous 95% guarantee?

**Q24.** Draft the smallest confirmatory protocol that identifies the D-minus-R contrast. Include allocation/dependence unit, disjoint discovery and held-out instances, frozen source/detector/checker versions, common budgets and feedback opportunity, shadow checks, success and safety gates, missing outcomes, and stopping rules. Explain the ceiling if the route adapters remain unavailable or the assisted arm A supplies solutions.

## Grading and advancement

Score each question from 0 to 4: 0, absent or wrong; 1, a label without support; 2, a partly correct calculation or argument; 3, correct result with its contract and evidence limit; 4, complete reasoning plus the requested counterexample or source binding. Maximum: 96.

Mastery requires at least 3 on every question and correction of every unsupported promotion of a finding, receipt, refusal, finite panel, or historical run. A high total cannot compensate for claiming a certificate from NO_NATIVE_PORT. This rubric evaluates the declared exercises; it grants no research or operational authority.